

PRAIRIE RIVER, AB, Canada

WMO: 715930

Lat: 58.627N

Lon: 111.679W

Elev: 212

StdP: 98.81

Time Zone: -7.00 (NAM)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-33.2	-40.2	0.1	-36.0	-37.0	0.1	-33.1	10.2	-15.4	9.1	-15.5	2.4	290	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	26.9	17.3	25.3	16.8	23.6	16.1	18.9	24.9	18.0	23.2	17.0	21.7	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.1	21.4	15.7	11.5	20.4	15.0	10.9	19.5	54.5	24.9	51.5	23.4	48.6	21.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.3	-39.1	30.5	2.7	1.5	-41.1	31.6	-42.6	32.4	-44.2	33.3	-46.1	34.3		
(5)			-39.2	21.1	2.7	1.5	-41.2	22.2	-42.7	23.0	-44.3	23.8	-46.2	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-18.6	-17.6	-11.4	-0.2	7.6	14.3	17.2	14.6	9.4	1.8	-8.0	-15.8	(6)
(7)		DBStd	14.21	9.17	7.37	8.18	6.67	5.48	3.99	2.90	3.80	4.24	5.07	6.49	8.03	(7)
(8)		HDD10.0	4411	885	772	663	307	108	9	0	7	62	258	539	801	(8)
(9)		HDD18.3	6898	1143	1005	921	555	334	129	56	125	269	512	789	1059	(9)
(10)		CDD10.0	596	0	0	0	2	34	139	223	149	43	4	0	0	(10)
(11)		CDD18.3	41	0	0	0	0	2	9	21	9	0	0	0	0	(11)
(12)		CDH23.3	399	0	0	0	0	27	105	179	83	4	1	0	0	(12)
(13)		CDH26.7	53	0	0	0	0	3	19	19	13	0	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.4	3.5	3.8	3.8	4.1	3.6	3.3	3.5	3.5	3.5	3.2	(14)	
(15)	Precipitation	PrecAvg	377	14	14	15	15	28	53	65	53	60	20	20	18	(15)
(16)		PrecMax	489	32	44	36	63	59	121	110	120	148	60	59	58	(16)
(17)		PrecMin	274	4	4	2	3	9	17	23	19	20	1	5	6	(17)
(18)		PrecStd	59	7	8	9	13	14	26	25	30	32	14	13	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	2.9	8.9	18.9	26.6	29.1	28.7	28.3	23.7	18.5	6.6	4.8	(19)
(20)			MCWB	2.4	0.4	4.1	9.9	14.4	17.4	18.0	19.6	16.4	12.5	3.4	2.6	(20)
(21)		2%	DB	1.0	-1.0	5.5	15.1	22.7	26.2	26.9	25.7	20.3	14.0	4.0	1.0	(21)
(22)			MCWB	-0.8	-2.4	2.0	7.5	12.4	15.0	17.7	18.6	14.6	9.0	1.6	-0.2	(22)
(23)		5%	DB	-2.0	-3.9	3.3	12.2	18.9	24.1	25.5	23.5	18.1	11.0	1.9	-1.6	(23)
(24)			MCWB	-2.9	-5.1	0.8	6.2	10.7	14.7	17.4	17.2	12.9	7.0	0.6	-2.3	(24)
(25)		10%	DB	-5.6	-6.8	1.0	9.2	16.4	21.9	23.8	21.6	16.1	9.0	0.5	-3.8	(25)
(26)			MCWB	-6.1	-7.4	-0.5	4.7	9.4	14.0	16.8	16.2	12.2	5.8	-0.3	-4.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	0.7	4.2	10.5	15.4	18.9	20.2	20.6	17.1	12.7	3.5	2.4	(27)
(28)			MCDWB	3.8	2.9	8.4	18.1	24.7	26.0	25.8	26.3	21.9	18.6	6.0	4.5	(28)
(29)		2%	WB	-0.7	-2.4	2.2	8.2	13.0	17.3	19.0	19.0	15.2	9.5	1.8	0.0	(29)
(30)			MCDWB	0.8	-0.9	5.1	13.9	19.9	22.6	24.9	24.8	19.4	12.1	3.3	0.8	(30)
(31)		5%	WB	-2.9	-5.0	1.0	6.5	11.5	15.9	18.2	17.7	13.8	7.4	0.7	-2.4	(31)
(32)			MCDWB	-1.9	-3.9	3.2	11.6	18.1	21.1	23.5	22.5	17.0	10.8	1.8	-1.6	(32)
(33)		10%	WB	-6.1	-7.4	-0.8	4.9	10.0	14.9	17.5	16.6	12.6	6.0	-0.4	-4.4	(33)
(34)			MCDWB	-5.6	-6.8	1.2	9.4	15.0	20.8	22.4	20.8	15.6	8.7	0.5	-3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	10.8	12.0	11.7	11.1	11.7	11.0	10.5	9.5	8.3	7.1	8.3	(35)
(36)			MCDBR	10.5	13.6	13.7	15.1	14.5	14.6	13.6	13.5	12.6	11.9	8.2	9.8	(36)
(37)			MCWBR	9.3	12.3	10.6	8.7	6.8	6.6	6.7	7.5	7.3	7.2	6.4	8.7	(37)
(38)		5% WB	MCDBR	10.7	13.5	13.0	14.4	13.0	12.0	12.1	12.8	11.2	11.2	7.8	9.5	(38)
(39)			MCWBR	9.5	12.2	10.2	8.5	6.4	6.0	6.3	7.3	6.8	7.1	6.3	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.219	0.244	0.284	0.327	0.348	0.370	0.391	0.355	0.318	0.286	0.234	0.197	(40)	
(41)		taud	2.215	2.307	2.273	2.362	2.412	2.351	2.360	2.435	2.542	2.545	2.294	2.117	(41)	
(42)		Ebn at Noon	651	815	865	883	885	865	835	844	832	764	625	544	(42)	
(43)		Edh at Noon	43	66	93	102	104	113	110	95	73	54	40	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.50	1.38	3.00	4.76	5.78	6.00	5.54	4.47	2.90	1.40	0.54	0.30	(44)	
(45)		RadStd	0.06	0.10	0.16	0.29	0.49	0.42	0.32	0.25	0.25	0.16	0.07	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page